

Project Proposal

Project Title: AstraTech: Development of a Multi-layered Autonomous Border Defense System (MABDS) with Integrated Non-Lethal Interdiction

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1. Executive Summary

Abstract

The effective surveillance and defense of India's diverse and expansive land borders present a persistent strategic challenge. Traditional methodologies, which rely heavily on static physical barriers and human patrolling, suffer from inherent limitations including high latency, vulnerability to environmental masking (foliage, weather), and the physiological limits of human endurance. These gaps often result in delayed detection of infiltration, leading to reactive rather than proactive engagement strategies.

AstraTech proposes the **Multi-layered Autonomous Border Defense System (MABDS)**, an integrated "Sense-and-Respond" ecosystem designed to automate intruder detection and provide an immediate, non-lethal interdiction capability. The system fuses data from two primary autonomous nodes: the **Tree-Mounted Advanced Intruder Sensing (T-MAIS)** unit, which provides aerial multi-spectral situational awareness, and the **Autonomous Self-Reliant Ground Defense Unit (ASGDU)**, a subterranean system utilizing seismic triangulation and a unique Multi-Charge CR Dispersal Canister (MC-CRDC).

By leveraging Edge AI for real-time threat discrimination and a secure LoRaWAN mesh for communication, MABDS aims to reduce false positives caused by wildlife by over 80% while ensuring a response time of under 10 seconds. This proposal outlines a phased implementation plan, prioritizing the immediate deployment of the detection grid followed by the integration of the non-lethal response mechanism, ensuring a scalable and ethical solution to modern border security challenges.

Key Objectives

- **Precision Detection:** Achieve >95% accuracy in distinguishing human intruders from local fauna using multi-modal sensor fusion (Thermal, Optical, Radar, Seismic)³³³³.

- **Zero-Latency Response:** Enable on-device Edge AI processing to detect, classify, and alert Command Control in <10 seconds, eliminating cloud-dependency latency.
- **Ethical Interdiction:** Deploy a "Human-in-the-Loop" authorized non-lethal response system (CR gas) to incapacitate intruders with a Minimal Effective Dose (MED), ensuring compliance with international norms.
- **Indigenous Reliability:** Design for 90% indigenous content, utilizing Indian manufacturing hubs for titanium fabrication and PCB assembly to support Atmanirbhar Bharat.

Relevance to DRDO

This project directly aligns with DRDO's mandate to equip the Indian Armed Forces with state-of-the-art technologies. Specifically, it addresses the critical need for "Smart Border Management" solutions that reduce physical risk to soldiers. The MABDS supports the Atmanirbhar Bharat initiative by reducing reliance on imported surveillance tech (e.g., foreign thermal cores) through the development of indigenous integration and algorithm capabilities, fostering a self-reliant defense ecosystem.

2. Introduction & Background

Problem Statement

India shares over 15,000 km of land borders, traversing terrain that includes the dense jungles of the Northeast, the salt marshes of Gujarat, and the high-altitude deserts of Ladakh.

Current border defense infrastructure is plagued by:

1. **High Latency:** Human patrols cover limited ground; a gap in patrol timing can leave sectors vulnerable for hours.
2. **Environmental Blind Spots:** Dense foliage and uneven terrain render traditional line-of-sight static cameras ineffective, creating "shadow zones" for infiltration.
3. **False Alarm Fatigue:** Legacy motion sensors trigger indiscriminately for wildlife (e.g., wild boars, cattle), causing "alert fatigue" where operators may ignore genuine warnings.
4. **Lack of Immediate Deterrence:** Detection systems are passive; they alert a remote team that may take minutes or hours to arrive, by which time the intrusion is complete.

Literature Review

Current solutions generally fall into two categories: unattended ground sensors (UGS) and aerial surveillance (drones/aerostats).

- **UGS Systems:** Conventional seismic sensors often lack the fidelity to distinguish between animal and human footsteps, leading to high false positive rates. Existing commercial solutions are often proprietary, expensive to import, and difficult to integrate with legacy Indian military networks.
- **Smart Fences:** Projects like the "Comprehensive Integrated Border Management System" (CIBMS) utilize thermal imagers and radar but are capital intensive and require significant power infrastructure, making them unsuitable for remote, off-grid areas.
- **Limitations:** Most research focuses purely on detection. There is a significant gap in integrated *response* systems—specifically, autonomous platforms capable of non-lethal interdiction without immediate human physical presence.

Project Scope

- **In Scope:**
 - Development of the T-MAIS Lite hardware and software stack for tree-mounted surveillance.
 - Fabrication of the ASGDU prototype, including the seismic sensor array and the MC-CRDC payload mechanism.
 - Implementation of Edge AI algorithms for human/animal classification.
 - Development of a secure, encrypted LoRaWAN communication protocol.
- **Out of Scope:**
 - Large-scale manufacturing (mass production).
 - Detection of high-altitude aerial threats (fighter jets, high-altitude drones).
 - Deep tunnel detection beyond 10 meters depth.

3. Proposed Solution & Technical Approach

System Architecture

The MABDS ecosystem operates on a tiered architecture:

1. **Tier 1: Sensing Layer (Distributed Nodes)**
 - **T-MAIS Lite:** Tree-mounted units providing thermal and optical data.
 - **ASGDU:** Buried units providing seismic and magnetic data.

2. Tier 2: Processing Layer (Edge Intelligence)

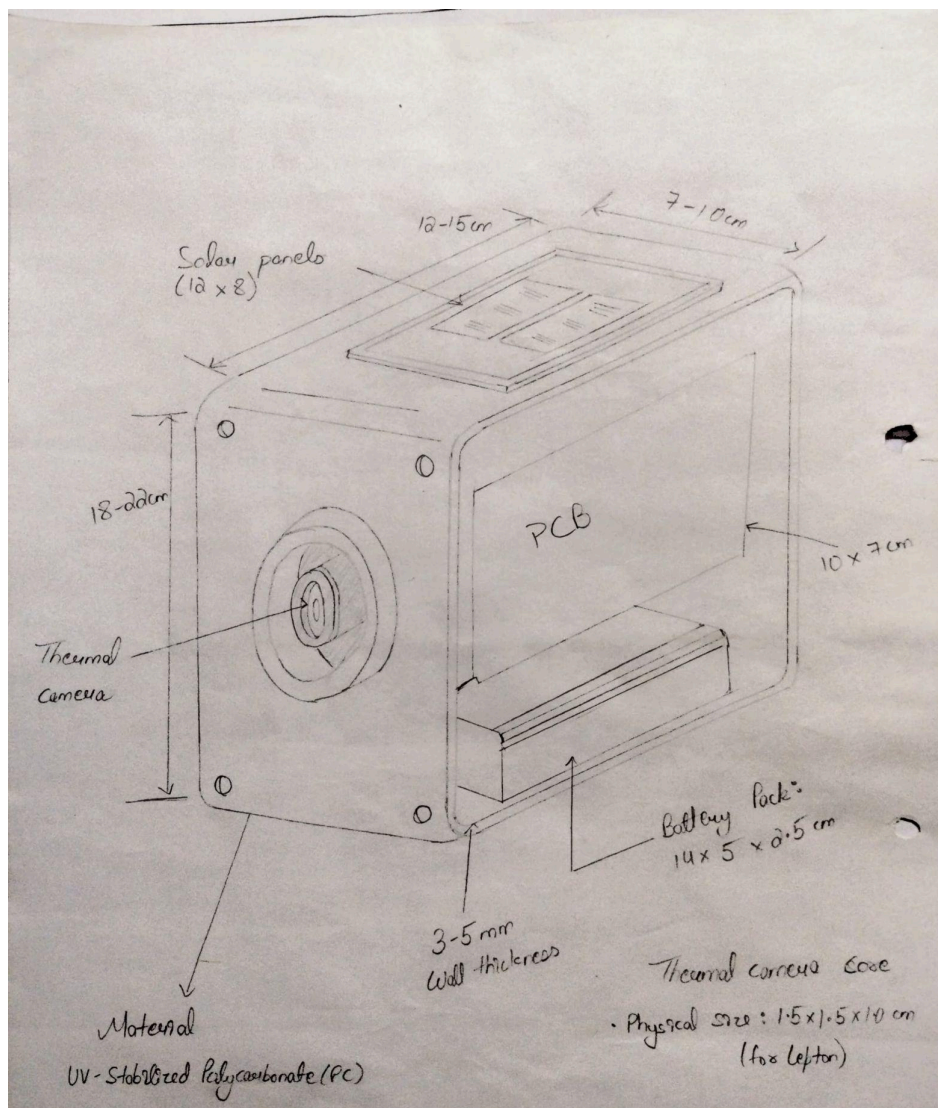
- Local NXP i.MX 8M Plus processors perform sensor fusion and classification on-device.

3. Tier 3: Communication Layer (Secure Mesh)

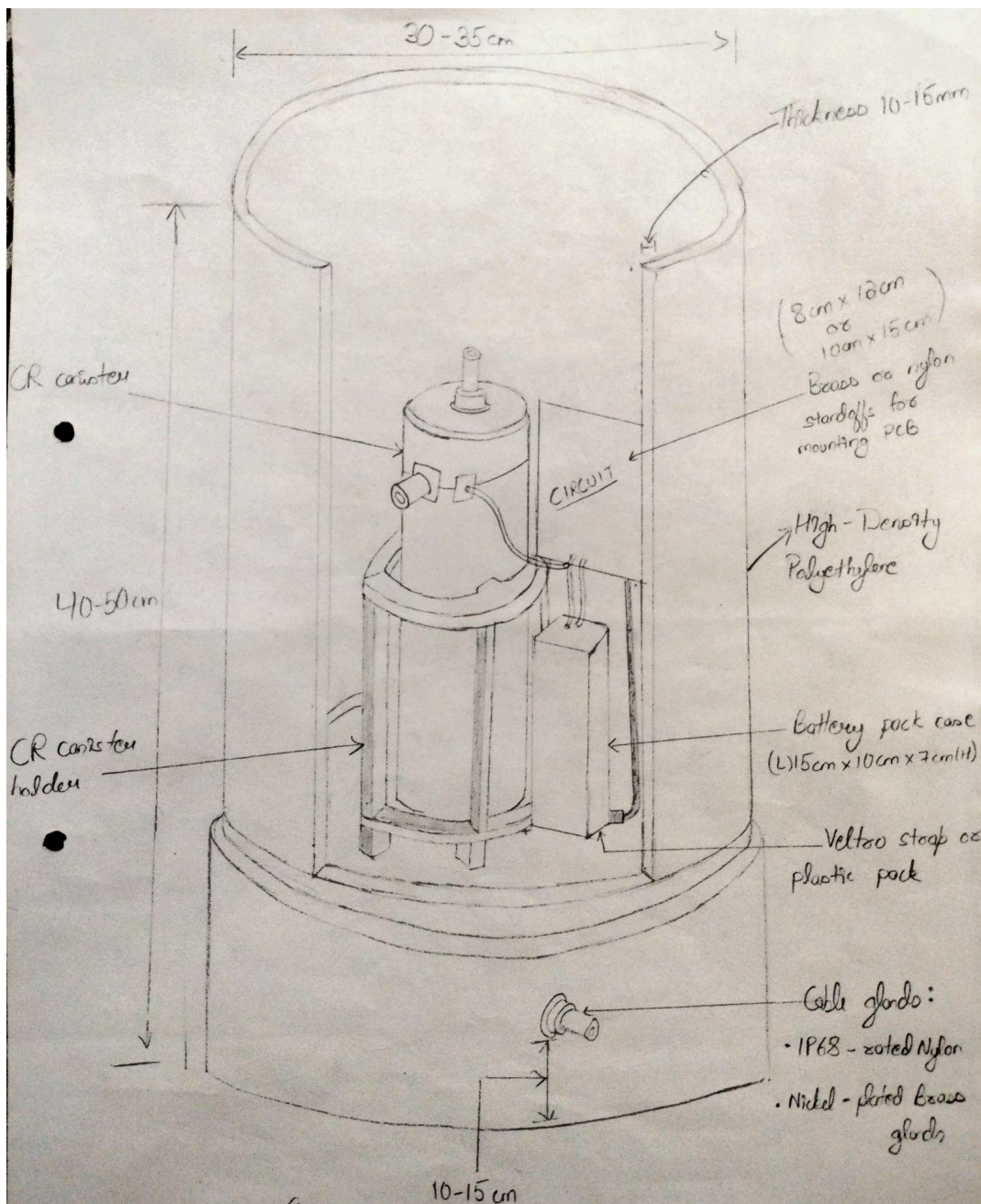
- LoRaWAN (Long Range Wide Area Network) for node-to-node and node-to-gateway communication.
- Satellite/Cellular backhaul for gateway-to-command center transmission.

4. Tier 4: Action Layer (Interdiction)

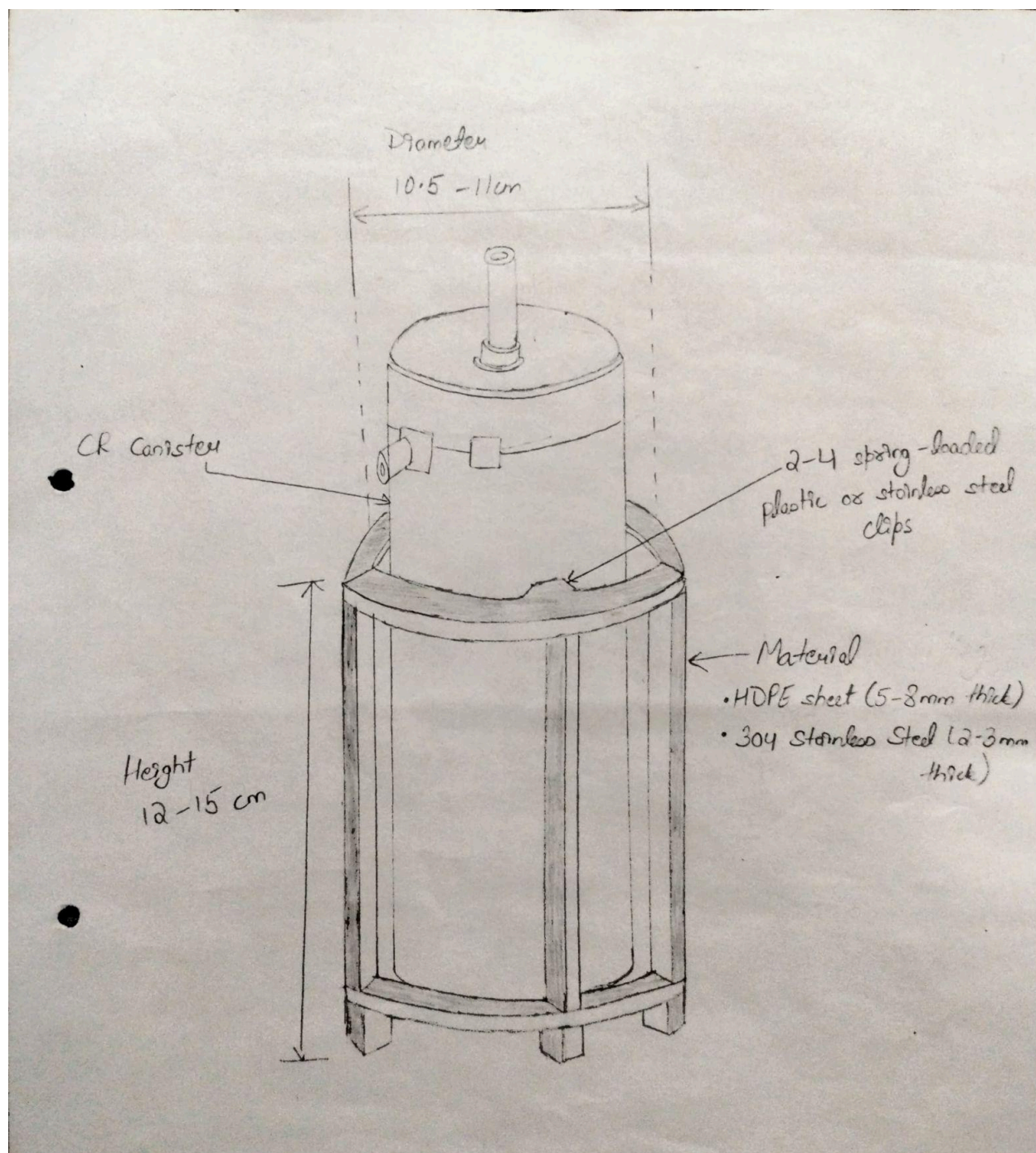
- MC-CRDC module for controlled gas release upon encrypted authorization.



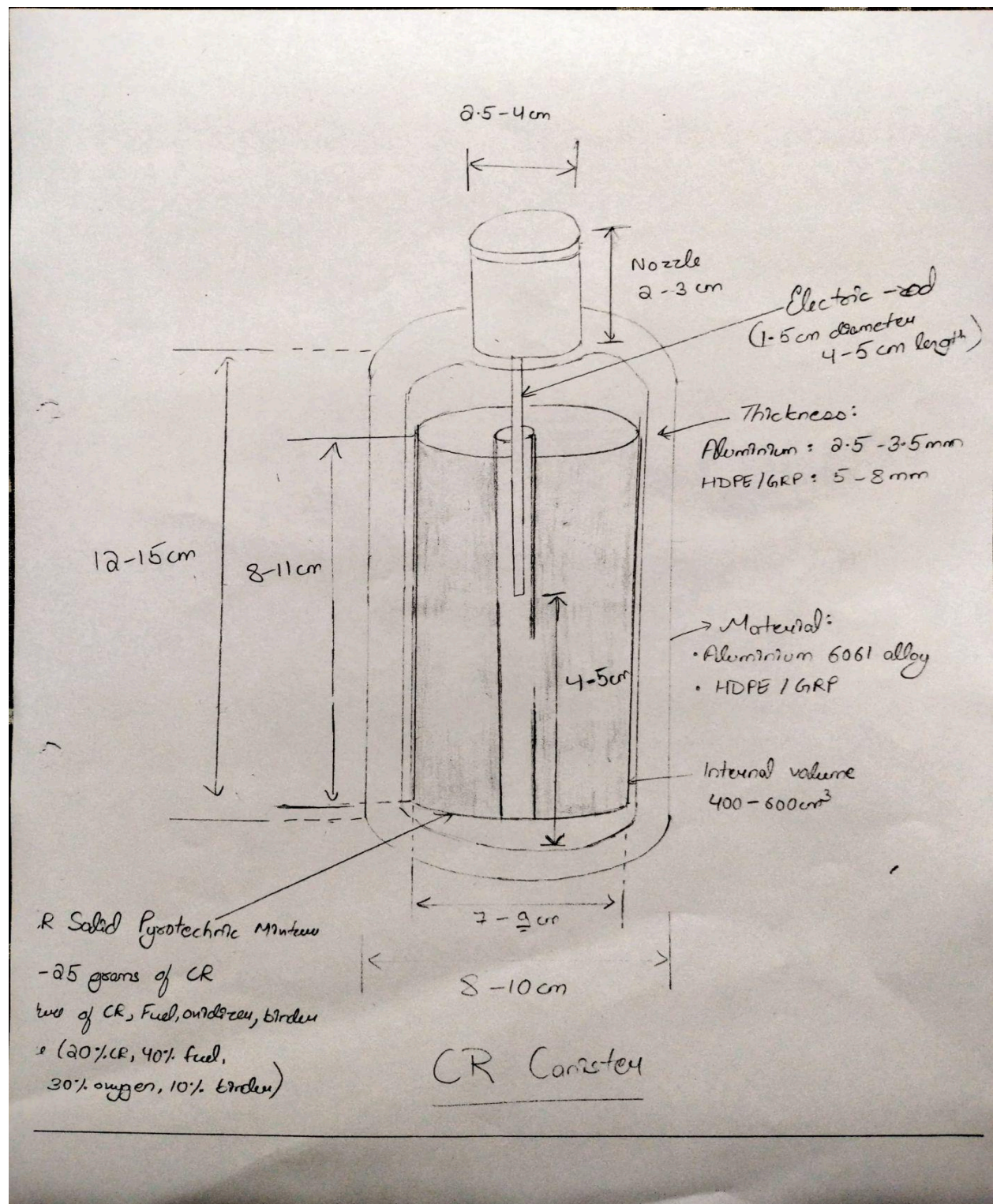
Tree-Mounting Device



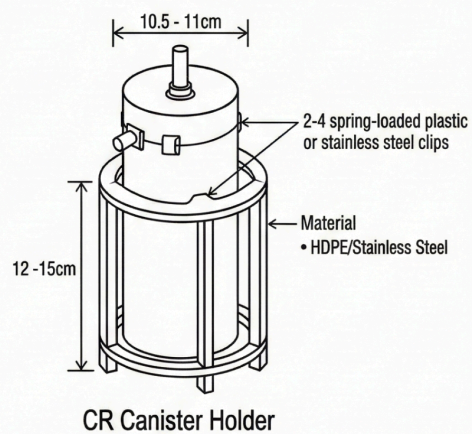
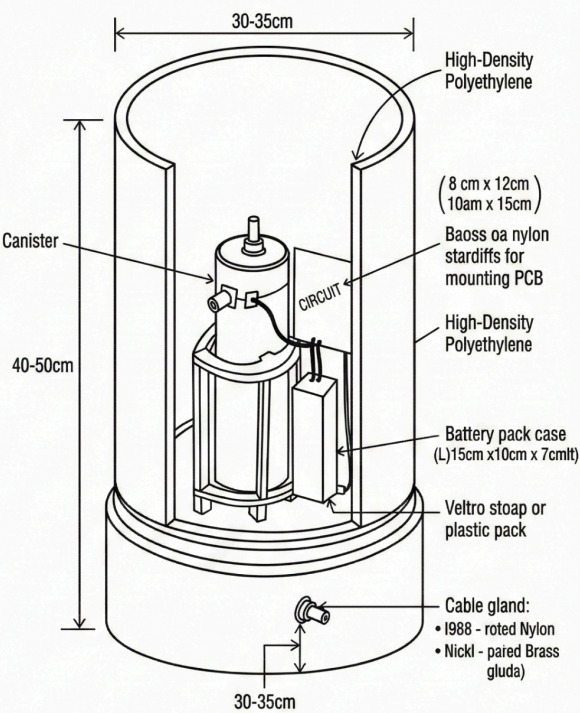
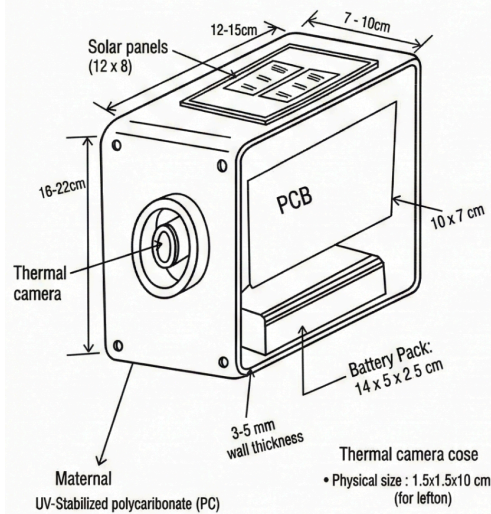
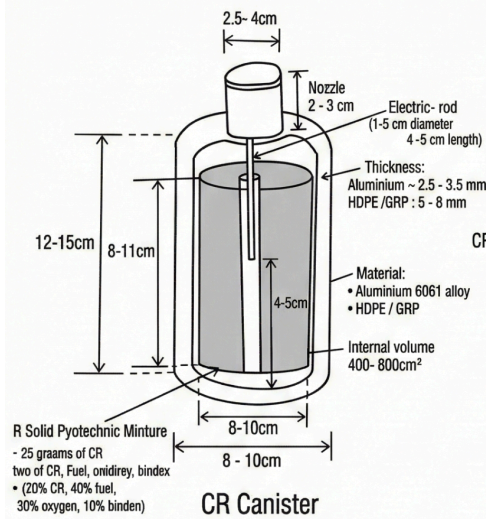
Outer-Shell Canister

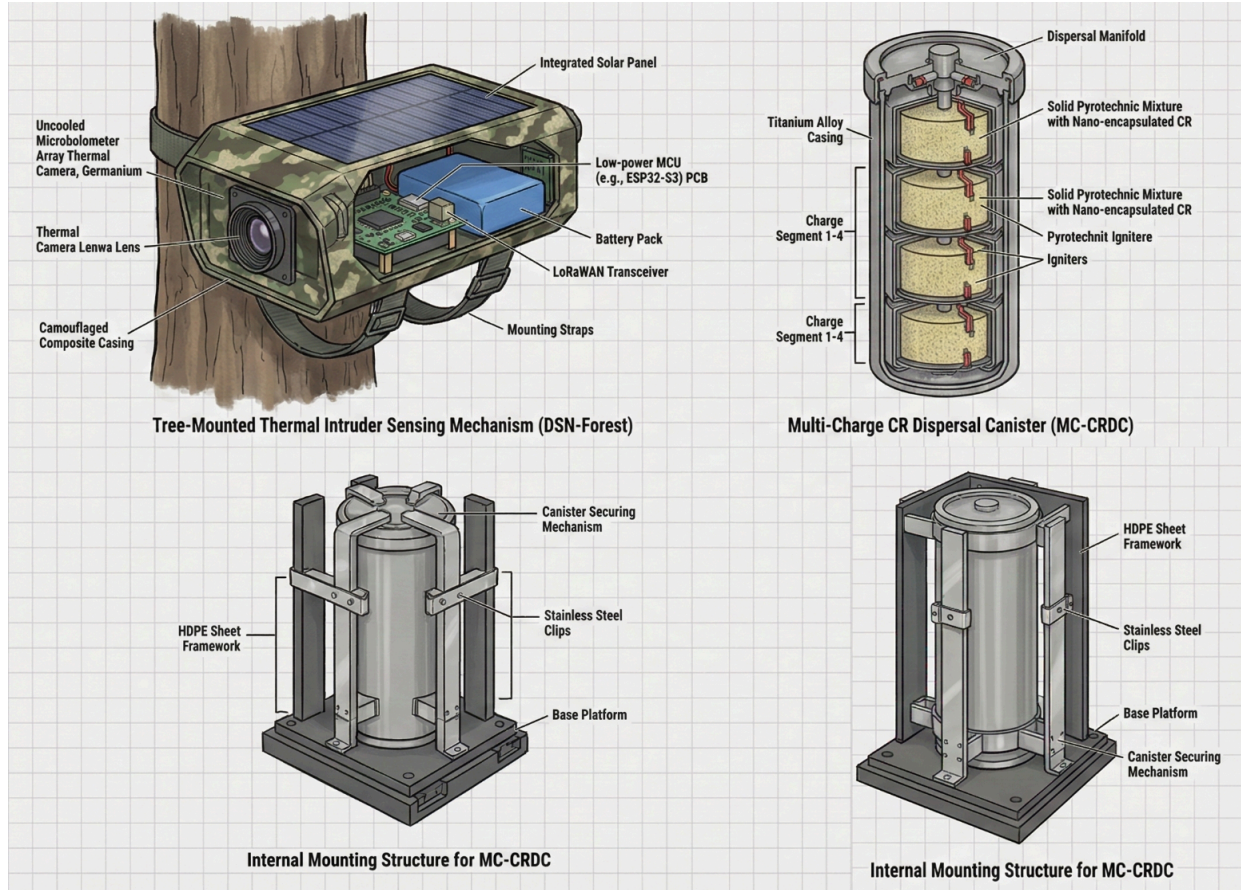


Internal CR Canister Mounting Mechanism



CR-Canister





Methodology

• Algorithms & Theory:

- Seismic Triangulation: The ASGDU utilizes Time Difference of Arrival (TDOA) across its 4 buried geophones to localize footsteps.

$$\Delta t_{(i,j)} = t_i - t_j = (1/v)(\|x - s_i\| - \|x - s_j\|)$$

Where,

$\Delta t_{(i,j)}$ — time difference between sensors i and j

v — wave velocity in soil

x — target location

s_i, s_j — sensor coordinates

- **Sensor Fusion (Kalman Filter):** Data from Thermal (z_t), Radar (z_r), and Seismic (z_s) sensors are fused using an Extended Kalman Filter (EKF) to estimate the target's

state vector (position, velocity), reducing uncertainty.

- **Convolutional Neural Networks (CNN):** A lightweight CNN (e.g., MobileNetV3) optimized for the NPU will classify thermal images. The loss function will prioritize minimizing false negatives (missing an intruder) over false positives.

- **Software Stack:**

- **Languages:** C++ (for sensor drivers/firmware), Python (for AI model training/inference).
- **Frameworks:** TensorFlow Lite (for Edge AI), ROS 2 (Robot Operating System) for internal module communication.
- **OS:** Yocto-based Linux for the main processor, FreeRTOS for the low-power microcontroller (ESP32).

- **Hardware Specifications:**

- **Processing Unit:** NXP i.MX 8M Plus (Quad-core Cortex-A53 @ 1.8GHz, 2.3 TOPS NPU).
- **Sensors:**
 - *Thermal:* Uncooled VOx Microbolometer (640x480, 19mm lens).
 - *Optical:* 1/2.8" Starlight CMOS (1920x1080).
 - *Seismic:* High-sensitivity Geophones (e.g., ADXL355 equivalent).
 - *Radar:* 24GHz FMCW module.
- **Actuators (MC-CRDC):**
 - *Igniters:* Low-voltage pyrotechnic squibs.
 - *Valves:* Rupture discs calibrated to 500-700 psi.
- **Materials:**
 - *Casing:* Titanium Alloy (Ti-6Al-4V) for the canister; Bio-mimetic composite for the T-MAIS shell.
 - *Insulation:* Ceramic fiber (alumina-silica).

Novelty

- **Integrated Non-Lethal Response:** This is the first proposed system to integrate a multi-shot, remote-controlled chemical dispersal mechanism (MC-CRDC) directly into an autonomous surveillance node, allowing for "Minimal Effective Dose" application.

- **Subterranean-Aerial Fusion:** The specific combination of tree-mounted thermal vision with buried seismic triangulation addresses the "foliage penetration" problem more effectively than either solution alone.

4. Implementation Plan

Phases of Development

- **Phase 1: Research & Design (Weeks 1–4)**
 - Finalize system requirement specifications (SRS).
 - Select specific sensors (thermal core, geophones) and procure development boards (NXP i.MX 8M Plus).
 - Design PCB schematics for the T-MAIS main control board.
 - Draft CAD models for the ASGDU casing and MC-CRDC internal manifold.
- **Phase 2: Prototyping & Simulation (Weeks 5–16)**
 - **Hardware:** Fabricate the T-MAIS Lite chassis using 3D printing (ABS) for fit checks. Assemble the first PCB revision.
 - **Software:** Train the initial AI model on open-source thermal datasets (e.g., FLIR ADAS) augmented with synthetic noise. Simulate seismic wave propagation in MATLAB to test triangulation algorithms.
 - **Payload:** Machine the dummy MC-CRDC canister (using Aluminum instead of Titanium for cost) to test the mechanical fit and rupture disc actuation (using compressed air).
- **Phase 3: Testing & Validation (Weeks 17–28)**
 - **Lab Testing:** Conduct climatic chamber tests (-20°C to +55°C) on the T-MAIS unit.
 - **Field Trials:** Deploy 2 T-MAIS units and 1 ASGDU (without live agent) in a controlled wooded area. Test detection range and LoRaWAN connectivity.
 - **Safety Testing:** Validate the "Human-in-the-Loop" software protocols to ensure the firing circuit *cannot* be triggered without the specific cryptographic key.
- **Phase 4: Documentation & Final Report (Weeks 29–32)**
 - Compile data logs from field trials (False Positive Rate, Detection Range).
 - Finalize the technical data package (TDP) with manufacturing blueprints.
 - Submit the final project report and research paper draft.

Gantt Chart (Text Representation)

- **Month 1:** Research, Component Selection, CAD Design.
- **Month 2-3:** PCB Fab, AI Model Training, Mechanical Prototyping.

- **Month 4-5:** Subsystem Assembly, Lab Bench Testing.
 - **Month 6-7:** Field Trials, Integration Debugging.
 - **Month 8:** Documentation, Final Reporting.
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5. Strategic Relevance & Applications

Military Application

- **Counter-Infiltration:** Deployed along the Line of Control (LoC), MABDS serves as a force multiplier, allowing a small team to monitor a vast frontage. The ASGDU provides an immediate option to stall insurgents, buying time for the Quick Reaction Team (QRT) to arrive.
- **Base Perimeter Security:** Protects remote forward operating bases (FOBs) from surprise attacks by providing a 400m early warning buffer and non-lethal denial of area.

Dual-Use Potential

- **Anti-Poaching:** The T-MAIS unit can be deployed in national parks (e.g., Kaziranga) to detect poachers. The AI can be retrained to track endangered species (rhinos/tigers) for conservation data while alerting rangers to human presence.
- **Critical Infrastructure Protection:** Securing perimeters of nuclear power plants or oil refineries against unauthorized entry.

Indigenization Impact

- **Atmanirbhar Bharat:** By developing the sensor fusion algorithms and the specialized MC-CRDC mechanism in India, the project reduces dependency on costly foreign perimeter defense systems (often Israeli or European).
 - **Local Supply Chain:** The project necessitates the development of local high-precision titanium machining and military-grade PCB assembly capabilities, strengthening the domestic defense industrial base.
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6. Resources & Requirements

Hardware/Software Needed

- **Computing:** High-performance workstation with NVIDIA GPU (RTX 3080 or higher) for training deep learning models.
- **Development Boards:** NXP i.MX 8M Plus EVK, ESP32-S3 modules, LoRaWAN Gateways (e.g., RakWireless).

- **Sensors:** Lepton 3.5 or equivalent Thermal Cores, Geophones (SM-24 or ADXL355), Radar modules.
- **Software Licenses:** MATLAB/Simulink (for seismic modeling), Altium Designer (for PCB layout), SolidWorks (for CAD).

Data Requirements

- Access to DRDO/Army datasets regarding "footstep acoustic signatures" in various Indian terrains (sand, snow, forest) would significantly accelerate the calibration of the seismic algorithms.
- Permissions to conduct controlled pyrotechnic testing (via PESO) for the squib ignition system.

7. Expected Deliverables

1. **Working Prototypes:**
 - 2 functional units of **T-MAIS Lite** (Tree-mounted detection) capable of thermal/optical streaming.
 - 1 functional unit of **ASGDU** (Ground defense) with seismic detection and a verified mechanical firing system (inert payload).
2. **Source Code & Models:**
 - Complete repository of C++/Python code for the Edge AI, Sensor Fusion, and Command Console.
 - Trained .tflite (TensorFlow Lite) models for human/animal classification.
3. **Schematics & Blueprints:**
 - Altium PCB project files for all custom boards.
 - CAD files (STEP/STL) for the ASGDU titanium casing and T-MAIS camouflaged shell.
4. **Final Technical Report:** A comprehensive document detailing the design methodology, test results (accuracy, range, latency), and user operating manual.

